

Tutorial Book



M.Sc Electrodynamics

D. T. McGuinness, Ph.D

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This TutorialBook is a
complement to it's corresponding Lecture.
When in doubt regarding theory, please consult the LectureBook.

List of Questions

1	Vector Calculus	1
1.1	Simple Vectors	1
1.2	Finding the Points	2
1.3	Vector Addition	2
1.4	Finding the Dot Product	2
1.5	Finding the Vector Properties	3
1.6	Find the Angle Between	3
1.7	Calculate the Vector Product	3
1.8	What is the Gradient	3
1.9	Double Integrals	3
1.10	Fluid Flow	4
1.11	Field Circulation	4
1.12	An Example of Divergence Theorem	4
1.13	Surface Area of an Implicit Surface	5
1.14	Stokes' Theorem Over a Hemisphere	5
1.15	The Volume of a Sphere	6
1.16	A Simple Dirac Integral	6
2	Electrostatics	7
2.1	Electric Field at a Distance	7
2.2	A Gaussian Sphere	7
2.3	Capacitance of Parallel Plates	7
2.4	Capacitance of Concentric Spheres	8
3	Advanced Mathematical Methods	9
3.1	2D Grounded Plates	9
3.2	A Long Rectangular Tube	10
4	Magnetostatics	12
4.1	Motion of a Cycloid	12
4.2	Lifting Weights	13
4.3	Finding the Current in a Wire	13
4.4	Magnetic Field at a Distance	13
4.5	Magnetic Field of a Long Solenoid	14

4.6	Magnetic Field at a Distance	15
5	Electrodynamics	16
5.1	An EMF Battery	16
5.2	Current Carrying Wire	16
5.3	The Induced E field of a Wire	17
5.4	A Coaxial Solenoid	17
5.5	The Self-Inductance of a Toroid	17
5.6	A Long Coaxial Cable	17

Vector Calculus

Simple Vectors Q1

Find the components of the vector $\mathbf{v}$ with given initial point P and terminal point Q . Find $|\mathbf{v}|$ and unit vector $\hat{\mathbf{v}}$.

$P(3, 2, 0)$,	$Q(5, -2, 2)$,
$P(1, 1, 1)$,	$Q(-4, -4, -4)$,
$P(1, 0, 1.2)$,	$Q(0, 0, 6.2)$,
$P(2, -2, 0)$,	$Q(0, 4, 6)$,
$P(4, 3, 2)$,	$Q(-4, -3, 2)$,
$P(0, 0, 0)$,	$Q(6, 8, 10)$.

Q1

The solution is as follows:

$$\begin{aligned}
 \mathbf{v} &= (5 - 3)\hat{x} + (-2 - 2)\hat{y} + (2 - 0)\hat{z} \\
 &= (2)\hat{x} + (-4)\hat{y} + (2)\hat{z}, \\
 |\mathbf{v}| &= \sqrt{(2)^2 + (-4)^2 + (2)^2} = 2\sqrt{6}. \\
 \hat{\mathbf{v}} &= \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \frac{(2)\hat{x} + (-4)\hat{y} + (2)\hat{z}}{2\sqrt{6}} \\
 &= \left(\frac{1}{\sqrt{6}}\right)\hat{x} + \left(-\frac{2}{\sqrt{6}}\right)\hat{y} + \left(\frac{1}{\sqrt{6}}\right)\hat{z} \quad \blacksquare \\
 \mathbf{v} &= (-4 - 1)\hat{x} + (-4 - 1)\hat{y} + (-4 - 1)\hat{z} \\
 &= (-5)\hat{x} + (-5)\hat{y} + (-5)\hat{z}, \\
 |\mathbf{v}| &= \sqrt{(-5)^2 + (-5)^2 + (-5)^2} = 5\sqrt{3}. \\
 \hat{\mathbf{v}} &= \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \frac{(-5)\hat{x} + (-5)\hat{y} + (-5)\hat{z}}{5\sqrt{3}} \\
 &= \left(-\frac{1}{\sqrt{3}}\right)\hat{x} + \left(-\frac{1}{\sqrt{3}}\right)\hat{y} + \left(-\frac{1}{\sqrt{3}}\right)\hat{z} \quad \blacksquare \\
 \mathbf{v} &= (0 - 1)\hat{x} + (0 - 0)\hat{y} + (6.2 - 1.2)\hat{z} \\
 &= (-1)\hat{x} + (0)\hat{y} + (5)\hat{z}, \\
 |\mathbf{v}| &= \sqrt{(-1)^2 + (0)^2 + (5)^2} = \sqrt{26}.
 \end{aligned}$$

$$\begin{aligned}
 \hat{\mathbf{v}} &= \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \frac{(-1)\hat{x} + (0)\hat{y} + (5)\hat{z}}{\sqrt{26}} \\
 &= \left(-\frac{1}{\sqrt{26}}\right)\hat{x} + (0)\hat{y} + \left(\frac{5}{\sqrt{26}}\right)\hat{z} \quad \blacksquare \\
 \mathbf{v} &= (0 - 2)\hat{x} + (4 - (-2))\hat{y} + (6 - 0)\hat{z} \\
 &= (-2)\hat{x} + (6)\hat{y} + (6)\hat{z}, \\
 |\mathbf{v}| &= \sqrt{(-2)^2 + (6)^2 + (6)^2} = 2\sqrt{19}. \\
 \hat{\mathbf{v}} &= \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \frac{(-2)\hat{x} + (6)\hat{y} + (6)\hat{z}}{2\sqrt{19}} \\
 &= \left(-\frac{1}{\sqrt{19}}\right)\hat{x} + \left(\frac{3}{\sqrt{19}}\right)\hat{y} + \left(\frac{3}{\sqrt{19}}\right)\hat{z} \quad \blacksquare \\
 \mathbf{v} &= (-4 - 4)\hat{x} + (-3 - 3)\hat{y} + (2 - 2)\hat{z} \\
 &= (-8)\hat{x} + (-6)\hat{y} + (0)\hat{z}, \\
 |\mathbf{v}| &= \sqrt{(-8)^2 + (-6)^2 + (0)^2} = 10. \\
 \hat{\mathbf{v}} &= \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \frac{(-6)\hat{x} + (-8)\hat{y} + (0)\hat{z}}{10} \\
 &= \left(-\frac{3}{5}\right)\hat{x} + \left(-\frac{4}{5}\right)\hat{y} + (0)\hat{z} \quad \blacksquare \\
 \mathbf{v} &= (6 - 0)\hat{x} + (8 - 0)\hat{y} + (10 - 0)\hat{z} \\
 &= (6)\hat{x} + (8)\hat{y} + (10)\hat{z}, \\
 |\mathbf{v}| &= \sqrt{(6)^2 + (8)^2 + (10)^2} = 10\sqrt{2}. \\
 \hat{\mathbf{v}} &= \frac{\mathbf{v}}{|\mathbf{v}|} \\
 &= \frac{(6)\hat{x} + (8)\hat{y} + (10)\hat{z}}{10\sqrt{2}} \\
 &= \left(\frac{3}{5\sqrt{2}}\right)\hat{x} + \left(\frac{4}{5\sqrt{2}}\right)\hat{y} + \left(\frac{1}{\sqrt{2}}\right)\hat{z} \quad \blacksquare
 \end{aligned}$$



Finding the Points Q2

Given the components of a vector $\mathbf{v} = [v_x, v_y, v_z]$ and a particular initial point P , find the corresponding terminal point Q and the length of $\mathbf{v}$ (i.e., $|\mathbf{v}|$).

$$\begin{array}{ll} \mathbf{v} = [3, -1, 0]; & P(4, 6, 0), \\ \mathbf{v} = [8, 4, 2]; & P(-8, -4, -2), \\ \mathbf{v} = [0.25, 2, 0.75]; & P(0, -0.5, 0), \\ \mathbf{v} = [3, 2, 6]; & P(4, 6, 0), \\ \mathbf{v} = [4, 2, -2]; & P(4, 6, 0), \\ \mathbf{v} = [3, -3, 3]; & P(4, 6, 0), \end{array}$$

Q2

Previously we have defined $\mathbf{v} = Q - P$. Here we have $\mathbf{v}$ and P . To calculate Q we only need to add individual components of the vector with the initial point P .

$$\begin{aligned} Q &= \mathbf{v} + P \\ &= (3+4)\hat{x} + (-1+6)\hat{y} + (0+0)\hat{z} \\ &= (7)\hat{x} + (5)\hat{y} + (0)\hat{z}, \\ |\mathbf{v}| &= \sqrt{(3)^2 + (-1)^2 + (0)^2} = \sqrt{10} \quad \blacksquare \\ Q &= \mathbf{v} + P \\ &= (8+(-8))\hat{x} + (4+(-4))\hat{y} + (-2+2)\hat{z} \\ &= (0)\hat{x} + (0)\hat{y} + (0)\hat{z}, \\ |\mathbf{v}| &= \sqrt{(8)^2 + (4)^2 + (2)^2} = 2\sqrt{21} \quad \blacksquare \\ Q &= \mathbf{v} + P \\ &= (0.25+0)\hat{x} + (2+(-0.5))\hat{y} + (0.75+0)\hat{z} \\ &= (0.25)\hat{x} + (1.5)\hat{y} + (0.75)\hat{z}, \\ |\mathbf{v}| &= \sqrt{(0.25)^2 + (1.5)^2 + (0.75)^2} = \sqrt{74}/4 \quad \blacksquare \\ Q &= \mathbf{v} + P \\ &= (3+4)\hat{x} + (2+6)\hat{y} + (6+0)\hat{z} \\ &= (7)\hat{x} + (8)\hat{y} + (6)\hat{z}, \\ |\mathbf{v}| &= \sqrt{(7)^2 + (8)^2 + (6)^2} = \sqrt{149} \quad \blacksquare \\ Q &= \mathbf{v} + P \\ &= (4+4)\hat{x} + (2+6)\hat{y} + (-2+0)\hat{z} \\ &= (8)\hat{x} + (8)\hat{y} + (-2)\hat{z}, \\ |\mathbf{v}| &= \sqrt{(8)^2 + (8)^2 + (-2)^2} = 2\sqrt{33} \quad \blacksquare \\ Q &= \mathbf{v} + P \\ &= (3+4)\hat{x} + (-3+6)\hat{y} + (3+0)\hat{z} \\ &= (7)\hat{x} + (3)\hat{y} + (3)\hat{z}, \end{aligned}$$

$$|\mathbf{v}| = \sqrt{(7)^2 + (3)^2 + (3)^2} = 2\sqrt{67} \quad \blacksquare$$

Vector Addition Q3

Let $\mathbf{a} = [2, 1, 0]$, $\mathbf{b} = [-4, 2, 5]$ and $\mathbf{c} = [0, 0, 3]$. Calculate the following vector operations:

$$\begin{array}{lll} 2\mathbf{a}, & -\mathbf{a}, & -1/2\mathbf{a}, \\ 5(\mathbf{a} - \mathbf{c}), & 5\mathbf{a} - 5\mathbf{c}, & (3\mathbf{a} - 5\mathbf{b}) + 2\mathbf{c}, \\ 3\mathbf{a} + (-5\mathbf{b} + 2\mathbf{c}), & \mathbf{a} + 2\mathbf{b}, & 2\mathbf{b} + \mathbf{a}. \end{array}$$

Q3

The answers are the following:

$$\begin{aligned} 2\mathbf{a} &= (4)\hat{x} + (2)\hat{y} + (0)\hat{z}, \\ -\mathbf{a} &= (-2)\hat{x} + (-1)\hat{y} + (0)\hat{z}, \\ -1/2\mathbf{a} &= (-1)\hat{x} + (-0.5)\hat{y} + (0)\hat{z}, \\ 5(\mathbf{a} - \mathbf{c}) &= (10)\hat{x} + (5)\hat{y} + (-15)\hat{z}, \\ 5\mathbf{a} - 5\mathbf{c} &= (10)\hat{x} + (5)\hat{y} + (-15)\hat{z}, \\ (3\mathbf{a} - 5\mathbf{b}) + 2\mathbf{c} &= (26)\hat{x} + (-7)\hat{y} + (-19)\hat{z}, \\ 3\mathbf{a} + (-5\mathbf{b} + 2\mathbf{c}) &= (26)\hat{x} + (-7)\hat{y} + (-19)\hat{z}, \\ \mathbf{a} + 2\mathbf{b} &= (-6)\hat{x} + (5)\hat{y} + (10)\hat{z}, \\ 2\mathbf{b} + \mathbf{a} &= (-6)\hat{x} + (5)\hat{y} + (10)\hat{z}. \end{aligned}$$

Finding the Dot Product Q4

Find the dot product (i.e., $\mathbf{a} \cdot \mathbf{b}$) on the lengths of $\mathbf{a} = [1, 2, 0]$ and $\mathbf{b} = [3, -2, 1]$ as well as the angle (θ) between vectors.

Q4

To find the angle between two vector we exploit the following identity:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= |\mathbf{a}||\mathbf{b}| \cos \theta, \\ \theta &= \cos^{-1} \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \right) \\ &= \cos^{-1} \left(\frac{3 + (-4) + 0}{\sqrt{(1)^2 + (2)^2 + (0)^2} \sqrt{(3)^2 + (-2)^2 + (1)^2}} \right), \\ &= \cos^{-1} \left(-\frac{1}{\sqrt{70}} \right) = 96.86 \text{ deg} \quad \blacksquare \end{aligned}$$

Finding the Vector Properties Q5

Let $\mathbf{a} = [2, 1, 4]$, $\mathbf{b} = [-4, 0, 3]$ and $\mathbf{c} = [3, -2, 1]$. Find the following descriptions.

$$|\mathbf{a}|, |\mathbf{b}|, |\mathbf{c}|,$$

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}), \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c},$$

$$\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a},$$

$$4\mathbf{a} \cdot 3\mathbf{c}, 12\mathbf{a} \cdot \mathbf{c},$$

$$|\mathbf{b} + \mathbf{c}|, |\mathbf{b}| + |\mathbf{c}|,$$

$$\mathbf{a} \cdot \mathbf{c}, |\mathbf{a}||\mathbf{c}|.$$

Q5

The answers are the following:

$$|\mathbf{a}| = \sqrt{21},$$

$$|\mathbf{b}| = 5,$$

$$|\mathbf{c}| = \sqrt{14},$$

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} = 12,$$

$$\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a} = 3,$$

$$4\mathbf{a} \cdot 3\mathbf{c} = 12\mathbf{a} \cdot \mathbf{c} = 96,$$

$$|\mathbf{b} + \mathbf{c}| = \sqrt{21},$$

$$|\mathbf{b}| + |\mathbf{c}| = 5 + \sqrt{14},$$

$$\mathbf{a} \cdot \mathbf{c} = 8,$$

$$|\mathbf{a}||\mathbf{c}| = 7\sqrt{6}.$$

Find the Angle Between Q6

Let $\mathbf{a} = [1, 1, 1]$, $\mathbf{b} = [2, 3, 1]$ and $\mathbf{c} = [-1, 1, 0]$.

Find the angle between the following:

$$(\mathbf{a} - \mathbf{c}) \text{ and } (\mathbf{b} - \mathbf{c}), \quad (\mathbf{a}) \text{ and } (\mathbf{b} - \mathbf{c}).$$

Q6

The answers are as follows:

$$33.21 \text{ deg}, \quad 22.20 \text{ deg} \quad \blacksquare$$

Calculate the Vector Product Q7

Find the vector product $\mathbf{a} \times \mathbf{b}$ of $\mathbf{a} = [1, 1, 0]$ and $\mathbf{b} = [3, 0, 0]$.

The solution is as follows:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & 1 & 0 \\ 3 & 0 & 0 \end{vmatrix} \\ &= (0) \hat{x} + (0) \hat{y} + (-3) \hat{z} \quad \blacksquare \end{aligned}$$

Q7What is the Gradient Q8

Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$.

Q8

The gradient is as follows:

$$\begin{aligned} \nabla r &= \nabla \sqrt{x^2 + y^2 + z^2} \\ &= \frac{x}{\sqrt{x^2 + y^2 + z^2}} \hat{x} \\ &\quad + \frac{y}{\sqrt{x^2 + y^2 + z^2}} \hat{y} + \frac{z}{\sqrt{x^2 + y^2 + z^2}} \hat{z} \quad \blacksquare \end{aligned}$$

Double Integrals Q9

Find the following double integrals:

$$\begin{aligned} \int_0^1 \int_x^{2x} (x+y)^2 dy dx, & \quad \int_0^1 \int_y^{\sqrt{y}} (1-2xy) dx dy, \\ \int_0^3 \int_x^3 \cosh(x+y) dy dx, & \quad \int_0^1 \int_0^{y^3} \exp y^4 dx dy. \end{aligned}$$

Q9

The solution to integrations are as follows:

$$\begin{aligned} &\int_0^1 \int_x^{2x} (x+y)^2 dy dx \\ &= \int_0^1 \int_x^{2x} x^2 + 2xy + y^2 dy dx, \\ &= \int_0^1 \left[xy^2 + xy^2 + \frac{y^3}{3} \right]_x^{2x} dx, \\ &= \int_0^1 \left(4x^3 + \frac{7x^3}{3} \right) dx, \\ &= \left[4x^3 + \frac{7x^4}{12} \right]_0^1 = \frac{19}{12} \quad \blacksquare \\ &\int_0^1 \int_y^{\sqrt{y}} (1-2xy) dx dy \\ &= \int_0^1 \left[x - x^2 y \right]_y^{\sqrt{y}} dy, \end{aligned}$$



$$\begin{aligned}
 &= \int_0^1 \left[(\sqrt{y} - y^2) - (y - y^3) \right] dy \\
 &= \int_0^1 \left[y^3 + \sqrt{y} - y^2 - y \right] dy, \\
 &= \left[\frac{y^4}{4} + \frac{2}{3} y^{3/2} - \frac{y^3}{3} - \frac{y^2}{2} \right] \Big|_0^1, \\
 &= \left(\frac{1}{4} + \frac{2}{3} - \frac{1}{3} - \frac{1}{2} \right) - (0) = \frac{1}{12} \quad \blacksquare \\
 &\int_0^3 \int_x^3 \cosh(x+y) dy dx \\
 &= \int_0^1 \left[\sinh(x+y) \right] \Big|_x^3 dx \\
 &= \int_0^1 [\sinh(3+x) - \sinh(2x)] dx
 \end{aligned}$$

Fluid Flow Q10

A fluid's velocity field is $\mathbf{F} = (x) \hat{x} + (z) \hat{y} + (y) \hat{z}$. Find the flow along the helix $\mathbf{l}(t) = (\cos t) \hat{x} + (\sin t) \hat{y} + (t) \hat{z}$ with a range of $0 \leq t \leq \pi/2$.

Q10

We first evaluate $\mathbf{F}$ on the curve:

$$\mathbf{F} = (x) \hat{x} + (z) \hat{y} + (y) \hat{z} = (\cos t) \hat{x} + (t) \hat{y} + (\sin t) \hat{z}$$

Substitute $x = \cos t$, $z = t$, $y = \sin t$.

and then find $d\mathbf{l}/dt$:

$$\frac{d\mathbf{l}}{dt} = (-\sin t) \hat{x} + (\cos t) \hat{y} + (0) \hat{z}.$$

Then we integrate $\mathbf{F} \cdot (d\mathbf{l}/dt)$ from $t = 0$ to $t = \pi/2$:

$$\begin{aligned}
 \mathbf{F} \cdot \frac{d\mathbf{l}}{dt} &= (\cos t)(-\sin t) + (t)(\cos t) + (\sin t)(0), \\
 &= -\sin t \cos t + t \cos t + \sin t.
 \end{aligned}$$

Which makes,

$$\begin{aligned}
 \text{Flow} &= \int_{t=a}^{t=b} \mathbf{F} \cdot \frac{d\mathbf{l}}{dt} dt \\
 &= \int_0^{\pi/2} (-\sin t \cos t + t \cos t + \sin t) dt, \\
 &= \left[\frac{\cos^2 t}{2} + t \sin t \right] \Big|_0^{\pi/2} \\
 &= \left(0 + \frac{\pi}{2} \right) - \left(\frac{1}{2} + 0 \right) = \frac{\pi}{2} - \frac{1}{2} \quad \blacksquare
 \end{aligned}$$

Field Circulation Q11

Find the circulation of the field $\mathbf{F} = (x-y) \hat{x} + x \hat{y}$ around the circle $\mathbf{l}(t) = (\cos t) \hat{x} + (\sin t) \hat{y} + (0) \hat{z}$ with a range of $0 \leq t \leq 2\pi$.

Q11

On the circle, $\mathbf{F} = (x-y) \hat{x} + (x) \hat{y} + (0) \hat{z} = (\cos t - \sin t) \hat{x} + (\cos t) \hat{y} + (0) \hat{z}$ and

$$\frac{d\mathbf{l}}{dt} = (-\sin t) \hat{x} + (\cos t) \hat{y} + (0) \hat{z}.$$

Then

$$\mathbf{F} \cdot \frac{d\mathbf{l}}{dt} = -\sin t \cos t + \underbrace{\sin^2 t + \cos^2 t}_1,$$

Gives.

$$\begin{aligned}
 \text{Circulation} &= \int_0^{2\pi} \mathbf{F} \cdot \frac{d\mathbf{l}}{dt} dt \\
 &= \int_0^{2\pi} (1 - \sin t \cos t) dt \\
 &= \left[t - \frac{\sin^2 t}{2} \right] \Big|_0^{2\pi} = 2\pi \quad \blacksquare
 \end{aligned}$$

An Example of Divergence Theorem Q12

Evaluate both sides of the Divergence theorem for the expanding vector field:

$$\mathbf{F} = (x) \hat{x} + (y) \hat{y} + (z) \hat{z},$$

over the sphere:

$$x^2 + y^2 + z^2 = a^2$$

Q12

The outer unit normal to $\mathcal{S}$, calculated from the gradient of $f(x, y, z) = x^2 + y^2 + z^2 - a^2$, is:

$$\begin{aligned}
 \hat{n} &= \frac{\nabla \mathcal{S}}{|\nabla \mathcal{S}|} \\
 &= \frac{(2x) \hat{x} + (2y) \hat{y} + (2z) \hat{z}}{\sqrt{4x^2 + 4y^2 + 4z^2}} \\
 &= \frac{(x) \hat{x} + (y) \hat{y} + (z) \hat{z}}{a}.
 \end{aligned}$$

where $x^2 + y^2 + z^2 = a^2$ on $\mathcal{S}$

Therefore:

$$(\mathbf{F} \cdot \hat{n}) da = \frac{x^2 + y^2 + z^2}{a} da = \frac{a^2}{a} da = a da.$$



This in turn gives us:

$$\begin{aligned} & \iint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) \, da \\ &= \iint_S a \, da = a \iint_S da \\ &= a (4\pi a^2) = 4\pi a^3. \end{aligned}$$

where Area of S is $4\pi a^2$

The divergence of $\mathbf{F}$ is:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 3,$$

So,

$$\begin{aligned} \iiint_V (\nabla \cdot \mathbf{F}) \, d\tau &= \iiint_V 3 \, d\tau \\ &= 3 \left(\frac{4}{3} \pi a^3 \right) = 4\pi a^3 \quad \blacksquare \end{aligned}$$

Surface Area of an Implicit Surface Q13

Find the area of the surface cut from the bottom of the paraboloid $x^2 + y^2 - z = 0$ by the plane $z = 4$.

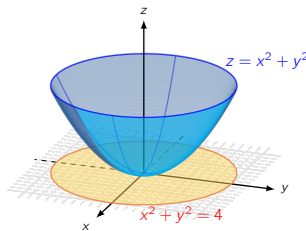


Figure 1.1: The paraboloid of "Surface Area of an Implicit Surface".

Q13

We sketch the surface S and the region R below it in the xy -plane (**Fig. 1.1**). The surface S is part of the level surface $F(x, y, z) = x^2 + y^2 - z = 0$, and R is the disk $x^2 + y^2 \leq 4$ in the xy -plane. To get a unit vector normal (i.e., $\hat{\mathbf{n}}$) to the plane R , we can take $\hat{\mathbf{n}} = \hat{\mathbf{z}}$. At any point (x, y, z) on the surface, we have:

$$\begin{aligned} F(x, y, z) &= x^2 + y^2 - z \\ \nabla F &= (2x) \hat{\mathbf{x}} + (2y) \hat{\mathbf{y}} + (-1) \hat{\mathbf{z}} \\ |\nabla F| &= \sqrt{(2x)^2 + (2y)^2 + (-1)^2} \\ &= \sqrt{4x^2 + 4y^2 + 1} \end{aligned}$$

$$|\nabla F \cdot \hat{\mathbf{n}}| = |\nabla F \cdot \hat{\mathbf{z}}| = |-1| = 1.$$

In the region R , the area is defined to be $dA = dx \, dy$. Therefore:

$$\begin{aligned} \text{Surface Area} &= \iint_R \frac{|\nabla F|}{|\nabla F \cdot \hat{\mathbf{n}}|} \, dA \\ &= \iint_{x^2+y^2 \leq 4} \sqrt{4x^2 + 4y^2 + 1} \, dx \, dy \\ &= \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} \, r \, dr \, d\theta \\ &= \int_0^{2\pi} \frac{1}{12} (4r^2 + 1)^{3/2} \Big|_0^2 \, d\theta \\ &= \int_0^{2\pi} \frac{1}{12} (17^{3/2} - 1) \, d\theta \\ &= \frac{\pi}{6} (17\sqrt{17} - 1) \quad \blacksquare \end{aligned}$$

Stokes' Theorem Over a Hemisphere Q14

Evaluate Stokes's theorem for the hemisphere S : $x^2 + y^2 + z^2 = 9$, $z \geq 0$, with its bounding circle C : $x^2 + y^2 = 9$, $z = 0$ and the field is:

$$\mathbf{F} = (y) \hat{\mathbf{x}} + (-x) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}.$$

Tip: Parametrisation of a circle is: $x = r \cos \theta$, $y = r \sin \theta$ and $da = 3/z \, dA$

Q14

We start by calculating the counter-clockwise circulation around C using the following parametrisation:

$$\begin{aligned} \ell(\theta) &= (3 \cos \theta) \hat{\mathbf{x}} + (3 \sin \theta) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}, \\ \text{where } 0 &\leq \theta \leq 2\pi. \end{aligned}$$

Using this we can calculate the counter-clockwise circulation.

$$\begin{aligned} d\ell &= (-3 \sin \theta \, d\theta) \hat{\mathbf{x}} + (3 \cos \theta \, d\theta) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}, \\ \mathbf{F} &= (y) \hat{\mathbf{x}} + (-x) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}} \\ &= (3 \sin \theta) \hat{\mathbf{x}} + (-3 \cos \theta) \hat{\mathbf{y}} + (0) \hat{\mathbf{z}}, \\ \mathbf{F} \cdot d\ell &= -9 \sin^2 \theta \, d\theta - 9 \cos^2 \theta \, d\theta = -9 \, d\theta, \\ \oint_C \mathbf{F} \cdot d\ell &= \int_0^{2\pi} -9 \, d\theta = -18\pi. \end{aligned}$$

For the curl of integral we have:

$$\begin{aligned} \nabla \times \mathbf{F} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_x & F_y & F_z \end{vmatrix} \\ &= (0 - 0) \hat{\mathbf{x}} + (0 - 0) \hat{\mathbf{y}} + (-1 - 1) \hat{\mathbf{z}} \end{aligned}$$



$$\begin{aligned}
 &= -2 \hat{z} \\
 \hat{n} &= \frac{\nabla S}{|\nabla S|} \\
 &= \frac{(x) \hat{x} + (y) \hat{y} + (z) \hat{z}}{\sqrt{x^2 + y^2 + z^2}} \\
 &= \frac{(x) \hat{x} + (y) \hat{y} + (z) \hat{z}}{3} \quad \text{Unit normal}
 \end{aligned}$$

Now it is time to define the area of integration (da):

$$\begin{aligned}
 da &= \frac{|\nabla S|}{|\nabla S \cdot \hat{z}|} dA \\
 &= \frac{|(2x) \hat{x} + (2y) \hat{y} + (2z) \hat{z}|}{2z} \\
 &= \frac{2 \sqrt{x^2 + y^2 + z^2}}{2z} \\
 &= \frac{3}{z} dA, \\
 \nabla \times \mathbf{F} \cdot \mathbf{n} da &= -\frac{2z}{3} \frac{3}{z} dA = -2 dA
 \end{aligned}$$

The cardinal direction $\hat{z}$ comes from being the direction **perpendicular** to the surface (S).

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} da = \iint_{x^2+y^2 \leq 9} -2 dA = -18\pi$$

The circulation around the circle equals the integral of the curl over the hemisphere ■

The Volume of a Sphere Q15

Find the volume of a sphere of radius R .

Q15

The derivation is as follows:

$$\begin{aligned}
 V &= \int d\tau \\
 &= \int_{r=0}^R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} r^2 \sin \theta dr d\theta d\phi, \\
 &= \left(\int_0^R r^2 dr \right) \left(\int_0^{\pi} \sin \theta d\theta \right) \left(\int_0^{2\pi} d\phi \right) \\
 &= \left(\frac{R^3}{3} \right) (2) (2\pi) = \frac{4}{3} \pi R^3 \quad \blacksquare
 \end{aligned}$$

A Simple Dirac Integral Q16

Evaluate the following integral:

$$\int_0^3 x^3 \delta(x-2) dx.$$

Q16

The delta function picks out the value of x^3 at the point $x = 2$, so the integral is $2^3 = 8$. Notice, however, that if the upper limit had been 1, instead of 3, the answer would be 0, because the spike would then be outside the domain of integration ■

2

Electrostatics

Electric Field at a Distance Q17

Find the $\mathbf{E}$ field a distance z above the midpoint between two (2) equal charges (q), a distance d apart:

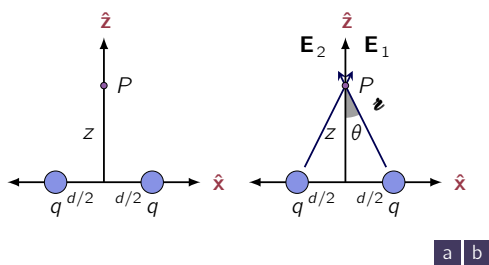


Figure 2.1: The diagram for the question “Electric Field at a Distance” (a) The geometrical description (b) Additional annotations added.

Q17

Let $\mathbf{E}_1$ be the field of the left charge, and $\mathbf{E}_2$ that of the right charge alone. Adding them, the horizontal components cancel and the vertical components add to:

$$E_z = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cos\theta.$$

Here $r = \sqrt{z^2 + (d/2)^2}$ and $\cos\theta = z/r$, therefore:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2qz}{[z^2 + (d/2)^2]^{3/2}} \hat{\mathbf{z}} \quad \blacksquare$$

A Gaussian Sphere Q18

Find the field outside a uniformly charged solid sphere of radius R and total charge q .

Q18

Imagine a spherical surface at radius $r > R$, shown below. This is called a Gaussian surface.

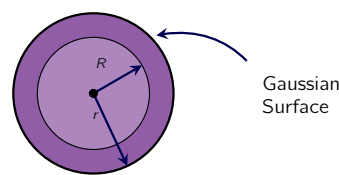


Figure 2.2: An imaginary Gaussian sphere.

Gauss's law says that:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}.$$

For this case it is $Q_{\text{enc}} = q$. At first glance this doesn't seem to get us very far, as the quantity we want (E) is buried inside the surface integral. Luckily, symmetry allows us to extract E from under the integral sign: E certainly points radially outward, as does $d\mathbf{a}$, so we can drop the dot product,

$$\int_S \mathbf{E} \cdot d\mathbf{a} = \int_S |\mathbf{E}| da$$

and the magnitude of E is constant over the Gaussian surface, so it comes outside the integral

$$\int_S |\mathbf{E}| da = |\mathbf{E}| \int_S da = |\mathbf{E}| 4\pi r^2$$

Therefore:

$$|\mathbf{E}| 4\pi r^2 = \frac{1}{\epsilon_0} q \quad \text{or} \quad \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \quad \blacksquare$$

Capacitance of Parallel Plates Q19

Find the capacitance of a parallel-plate capacitor consisting of two (2) metal surfaces of area A held a distance d apart.



Q19

If we were to put $+Q$ on the top and Q on the bottom, they will spread out uniformly over the two (2) surfaces, provided the area is reasonably large and the separation small. The surface charge density, then, is $\sigma = Q/A$ on the top plate, and so the field, is:

$$\frac{1}{\epsilon} \frac{Q}{A}$$

The potential difference between the plates is therefore

$$V = \frac{Q}{A\epsilon_0} d \quad \text{therefore} \quad C = \frac{A\epsilon_0}{d} \quad \blacksquare$$

Capacitance of Concentric Spheres ——— Q20

Find the capacitance of two (2) concentric spherical metal shells, with radii a and b .

Q20

Let's place the charge $+Q$ on the inner sphere, and Q on the outer one. The field between the spheres is:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$$

so the potential difference between them is:

$$\begin{aligned} V &= - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \frac{1}{4\pi\epsilon_0} Q \int_a^b \frac{1}{r^2} dr \\ &= \frac{1}{4\pi\epsilon_0} Q \left(\frac{1}{a} - \frac{1}{b} \right) \end{aligned}$$

Given V is proportional to Q , the capacitance is:

$$C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{ab}{(b-a)} \quad \blacksquare$$

3

Advanced Mathematical Methods

2D Grounded Plates Q21

Two infinitely long grounded metal plates at $y = 0$ and $y = a$ are connected at $x = \pm b$ by metal strips maintained at a constant potential V_0 as shown. A thin layer of insulation at each corner prevents the metal plates from shorting out. Find the potential inside the resulting rectangular pipe.

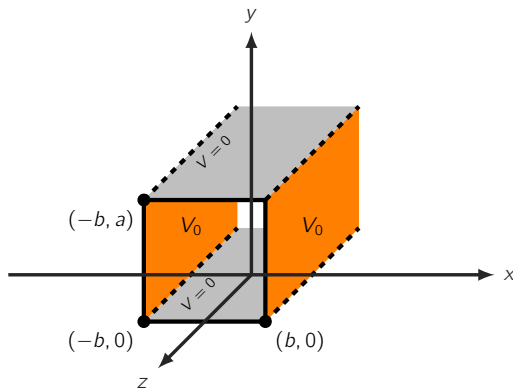


Figure 3.1: Diagram for the question "2D Grounded Plates".

Q21

As can be seen from the figure, the configuration is independent from z -direction. Therefore, our goal here is to solve a 2D Laplace's equation ($\nabla^2 f = 0$) is as follows:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0,$$

which are subject to the following boundary conditions:

- i. $V = 0$ when $y = 0$
- ii. $V = 0$ when $y = a$
- iii. $V = V_0(y)$ when $x = \pm b$
- iv. $V \rightarrow 0$ as $x \rightarrow \pm \infty$

The fourth option comes not from the mathematical description of the problem, but from physical

intuition. To solve this problem we need to use Separation of Variables method which is as follows. Assume a solution with the following description,

$$V(x, y) = X(x)Y(y).$$

Putting this into the original Laplace equation gives:

$$Y(y)\frac{d^2 X}{dx^2} + X(x)\frac{d^2 Y}{dy^2} = 0.$$

It is time to isolate the variables,

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0.$$

First term depends on x , and the second term depends on y . From this we can write:

$$f(x) + g(y) = 0.$$

For this to be true both of these functions needs to be constants.

$$\frac{d^2 X}{dx^2} = k^2 X, \quad \frac{d^2 Y}{dy^2} = -k^2 Y.$$

It is time to solve these equations, let's have a look at the first one:

$$\frac{d^2 X}{dx^2} = k^2 X.$$

In ODE this is classified as an homogeneous 2nd order linear differential equation with the generic form of

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + c y(x) = 0.$$

where a, b, c are arbitrary constants. The general solution for these kind of equation is,

$$y(x) = A \exp -r_1 x + B \exp -r_2 x$$

where

$$r_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}.$$

Based on this template, the solution to the first ODE is;

$$X(x) = A \exp kx + B \exp -kx,$$



If we were to apply the second ODE to the previously mentioned solution template we arrive at the following.

$$Y(y) = C \exp -jy + D \exp +jy$$

or

$$Y(y) = C \sin ky + D \cos ky.$$

While the imaginary solution is perfectly valid, we want to use the real only solution as the equation we are trying to solve is real as well. But let's continue and add the derived solutions together:

$$\begin{aligned} V(x, y) &= X(x) Y(y) \\ &= (A \exp kx + B \exp -kx) \\ &\quad \times (C \sin ky + D \cos ky) \end{aligned}$$

Here, however, we cannot set $A = 0$ as the region in question does not extend to $x = \infty$, so $\exp kx$ is perfectly acceptable, whereas, the situation is symmetric with respect to x , so $V(-x, y) = V(x, y)$ and it follows $A = B$. Using the hyperbolic cosine function identity:

$$A \exp kx + A \exp -kx = 2A \cosh kx,$$

distributing A to C and D , we have;

$$V(x, y) = \cosh kx (2AC \sin ky + 2AD \cos ky)$$

Boundary conditions **(i)** and **(ii)** require D to be equal to zero and $k = n\pi/a$ so:

$$V(x, y) = 2AC \cosh (n\pi x/a) \sin (n\pi y/a),$$

Because $V(x, y)$ is even in x , it will automatically meet condition **(iv)** if it fits **(iii)**. It remains, therefore, to construct the linear combination,

$$V(x, y) = \sum_{n=1}^{\infty} G_n \cosh (n\pi x/a) \sin (n\pi y/a),$$

where $G = 2AC$.

and pick coefficients C_n such that it satisfies the condition on **(iii)**:

$$V(b, y) = \sum_{n=1}^{\infty} G_n \cosh (n\pi b/a) \sin (n\pi y/a) = V_0,$$

A Long Rectangular Tube Q22

An infinitely long rectangular metal pipe (sides a and b) is grounded, but one end at $x = 0$, is maintained at a specific potential $V_0(y, z)$ as indicated above. Find the potential inside the pipe.

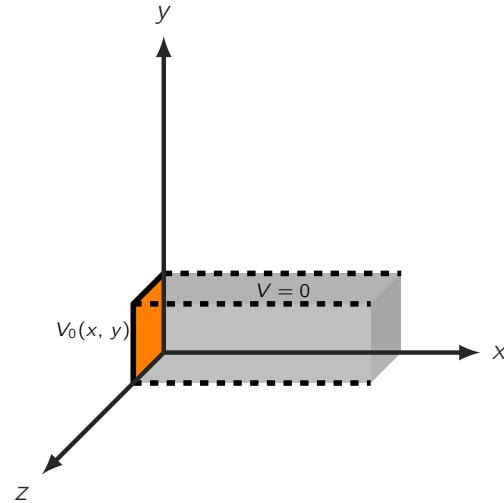


Figure 3.2

Q22

This is a 3D problem with the following **Laplace** equation:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0,$$

which are subject to the following boundary conditions given by the question either explicitly or implicitly:

- (i)** $V = 0$ when $y = 0$,
- (ii)** $V = 0$ when $y = a$,
- (iii)** $V = 0$ when $z = 0$,
- (iv)** $V = 0$ when $z = b$,
- (v)** $V \rightarrow 0$ as $x \rightarrow \infty$,
- (vi)** $V = V_0(x, y)$ when $x = 0$.

As usual, we are looking for solutions that are products using **separation of variables**:

$$V(x, y, z) = X(x) Y(y) Z(z)$$

Putting this into the Laplace equation presents to us:

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0.$$



Where once we isolate the variables, we can get the following relation:

$$\begin{aligned}\frac{1}{X} \frac{d^2 X}{dx^2} &= C_1, \\ \frac{1}{Y} \frac{d^2 Y}{dy^2} &= C_2, \\ \frac{1}{Z} \frac{d^2 Z}{dz^2} &= C_3, \quad \text{with} \\ C_1 + C_2 + C_3 &= 0.\end{aligned}$$

To keep this solution relatively short, We will assume C_1 **must** be positive and C_2, C_3 are negative. Setting $C_2 = -k^2$ and $C_3 = -l^2$, we have $C_1 = k^2 + l^2$, and therefore:

$$\begin{aligned}\frac{d^2 X}{dx^2} &= (k^2 + l^2) X, \\ \frac{d^2 Y}{dy^2} &= -k^2 Y, \\ \frac{d^2 Z}{dz^2} &= -l^2 Z\end{aligned}$$

We have yet again turned a Partial Differential Equation (PDE) into a series of Ordinary Differential Equation (ODE)s. The solutions are:

$$\begin{aligned}X(x) &= A \exp\left(\sqrt{k^2 + l^2}x\right) + B\left(-\sqrt{k^2 + l^2}x\right), \\ Y(y) &= C \sin ky + D \cos ky, \\ Z(z) &= E \sin lz + F \cos lz.\end{aligned}$$

Boundary condition **(v)** implies $A = 0$, **(i)** gives $D = 0$ and **(iii)** yields $F = 0$, whereas **(ii)** and **(iv)** require $k = n\pi/a$ and $l = m\pi/b$, where n and m are positive integers. Combining the remaining constants, we are presented with:

$$\begin{aligned}V(x, y, z) &= \\ &C \exp\left(-\pi \sqrt{(n/a)^2 + (m/b)^2} x\right) \\ &\times \sin(n\pi y/a) \sin(m\pi z/b).\end{aligned}$$

Magnetostatics

Motion of a Cycloid Q23

An unusual trajectory occurs if we were to include a **uniform electric field**, at right angles to the magnetic one. Suppose, for instance, that $\mathbf{B}$ points in the $\hat{x}$ -direction, and $\mathbf{E}$ in the $\hat{z}$ -direction, as shown in Fig. 5.7.

A positive charge is released from the origin. If then, what path will it follow?

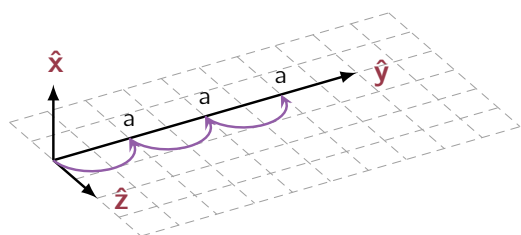


Figure 4.1: The diagram for the question "Motion of a Cycloid".

Q23

Let's think it through qualitatively, first. Initially, the particle is at rest, so the magnetic force is zero, and the electric field ($\mathbf{E}$) accelerates the charge in the $\hat{z}$ -direction.

As it picks up speed, a magnetic force develops which, pulls the charge around to the right. The faster it goes, the stronger F_{mag} becomes; eventually, it curves the particle back around towards the $\hat{y}$ -axis. At this point the charge is moving **against** the electrical force, so it begins to slow down and the magnetic force then decreases, the electrical force takes over, which brings the particle to rest at point a , shown in Fig. 4.1. There the entire process starts again, carrying the particle over to point b , and so on.

Now let's explain this mathematically. There being no force in the $\hat{x}$ -direction, the position of the particle at any time t can be described by the vector $(0, y(t), z(t))$; the velocity is therefore

$$\mathbf{v} = [0, \dot{y}, \dot{z}]$$

where dots indicate time derivatives. Therefore

$$\nabla \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & \dot{y} & \dot{z} \\ B & 0 & 0 \end{vmatrix} = B \frac{dz}{dt} \hat{y} - B \frac{dy}{dt} \hat{z}$$

and hence, applying Newton's 2nd law, we get

$$\begin{aligned} \mathbf{F} &= Q(\mathbf{E} + \nabla \times \mathbf{B}) \\ &= Q(E\hat{z} + B\dot{z}\hat{y} - B\dot{y}\hat{z}) \\ &= m\mathbf{a} = m(\ddot{y}\hat{y} + \ddot{z}\hat{z}). \end{aligned}$$

Or, treating the $\hat{y}$ and $\hat{z}$ components separately,

$$QB\dot{z} = m\ddot{y}, \quad QE - QB\dot{y} = m\ddot{z}.$$

For convenience, let

$$\omega \equiv \frac{QB}{m}. \quad (4.1)$$

This is the **cyclotron frequency**, at which the particle would revolve in the absence of any electric field. Then the equations of motion take the form

$$\ddot{y} = \omega\dot{z} \quad \text{and} \quad \ddot{z} = \omega\left(\frac{E}{B} - \dot{y}\right). \quad (4.2)$$

Their general solution is:

$$\begin{aligned} y(t) &= C_1 \cos \omega t + C_2 \sin \omega t + (E/B)t + C_3, \\ z(t) &= C_2 \cos \omega t - C_1 \sin \omega t + C_4. \end{aligned}$$

As **coupled differential equations**, they are easily solved by differentiating the first and using the second to eliminate $\ddot{z}$. But the particle started from rest ($\dot{y}(0) = \dot{z}(0) = 0$), at the origin ($y(0) = z(0) = 0$); these four conditions determine the constants C_1 , C_2 , C_3 , and C_4 :

$$y(t) = \frac{E}{\omega R}(\omega t - \sin \omega t),$$



$$z(t) = \frac{E}{\omega R}(1 - \cos \omega t).$$

In this form, the answer is **NOT** very clear, but if we let,

$$R \equiv \frac{E}{\omega B}, \quad (4.3)$$

and eliminate the sines and cosines by exploiting the trigonometric identity $\sin^2 \omega t + \cos^2 \omega t = 1$, we find that

$$(y - R\omega t)^2 + (z - R)^2 = R^2$$

This is the formula for a **circle**, of radius R , whose center $(0, R\omega t, R)$ travels in the $\hat{y}$ -direction at a constant speed

$$u = \omega R = \frac{E}{B} \quad (4.4)$$

The particle moves as though it were a spot on the rim of a wheel rolling along the y axis. The curve generated in this way is called a cycloid. Notice that the overall motion is **not** in the direction of E , as you might suppose, but perpendicular to it.

Lifting Weights Q24

A rectangular loop of wire, supporting a mass of m , hangs **vertically** with one end in a uniform magnetic field $\mathbf{B}$, which points into the page. For what current value of I , in the loop, would the magnetic force upward exactly balance the gravitational force downward?

Q24

First of all, the current **must circulate clockwise**, for $(\mathbf{I} \times \mathbf{B})$ in the horizontal segment to point upward. The force is

$$F_{\text{mag}} = IBa$$

where a is the width of the loop. An important note to mention is that the magnetic forces on the two vertical segments cancel. For F_{mag} to balance the weight (mg), we must therefore have:

$$I = \frac{mg}{Ba} \quad (4.5)$$

And if this current were to be applied, the mass would be there in mid-air.

However, What happens if we were to **increase** the current? Then the upward magnetic force exceeds the downward force of gravity, and the loop rises, lifting the weight. Somebody's doing work, and it

sure looks as though the magnetic force is responsible.

Indeed, we can write

$$W_{\text{mag}} = F_{\text{mag}}h = IBah, \quad (4.6)$$

where h is the distance the loop rises.

Finding the Current in a Wire Q25

A current I is uniformly distributed over a wire of circular cross section, with radius a .

- Please find the volume current density (J).
- Suppose the current density in the wire is proportional to the distance from the axis,

$$J = ks \quad \text{for some constant } k.$$

Find the total current in the wire.

Q25

The solutions are as follows:

- The area (perpendicular to the flow) is πa^2 , so

$$J = \frac{I}{\pi a^2}.$$

This was straightforward given the **current density was uniform**.

- Given J varies with s , we need to do **integration**. The current through the a small area of the cross-section is $J da_{\perp}$, and $da_{\perp} = s ds d\phi$. Therefore:

$$\begin{aligned} I &= \int (ks) (s ds d\phi) \\ &= 2\pi k \int_0^a s^2 ds = \frac{2\pi k a^3}{3} \quad \blacksquare \end{aligned}$$

Magnetic Field at a Distance Q26

Find the magnetic field a distance s from a long straight wire carrying a steady current I (Fig. 5.18). dI'

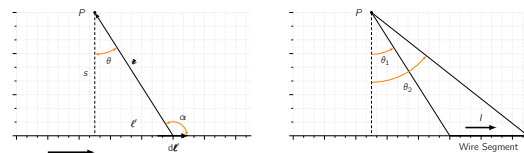


Figure 4.2: Diagrams for the question "Magnetic Field at a Distance."



Q26

In the diagram, $(d\ell' \times \hat{z})$ points **out** of the page, and has the magnitude

$$d\ell' \sin \alpha = d\ell' \cos \theta$$

Also, $\ell' = s \tan \theta$, so

$$d\ell' = \frac{s}{\cos^2 \theta} d\theta$$

and $s = z \cos \theta$, so

$$\frac{1}{z^2} = \frac{\cos^2 \theta}{s^2}$$

Therefore:

$$\begin{aligned} B &= \frac{\mu_0 I}{4\pi} \int_{\theta_1}^{\theta_2} \left(\frac{\cos^2 \theta}{s^2} \right) \left(\frac{s}{\cos^2 \theta} \right) \cos \theta d\theta \\ &= \frac{\mu_0 I}{4\pi s} \int_{\theta_1}^{\theta_2} \cos \theta d\theta = \frac{\mu_0 I}{4\pi s} (\sin \theta_2 - \sin \theta_1). \end{aligned}$$

The aforementioned equation gives the field of any **straight segment of wire**, in terms of the initial and final angles θ_1 and θ_2 . Of course, a finite segment by itself could never support a steady current, but it might be a **piece** of some closed circuit, and this equation would then represent its contribution to the total field. In the case of an **infinite** wire, $\theta_1 = -\pi/2$ and $\theta_2 = \pi/2$, so we obtain

$$B = \frac{\mu_0 I}{2\pi s}.$$

Please observe that the field is **inversely proportional to the distance** from the wire—just like the electric field of an infinite line charge. In the region below the wire, B points into the page, and in general, it “circles around” the wire, in accordance with the right-hand rule:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}.$$

Magnetic Field of a Long Solenoid — Q27

Find the magnetic field of a **very long solenoid**, consisting of n closely wound turns per unit length on a cylinder of radius R , each carrying a steady current of I .

Q27

First of all, we need to determine the direction of B . Could it have a radial component? No. For if B_s were **positive**; when we reversed the direction of the current, B_s would then become **negative**. However, switching I is physically equivalent to turning the solenoid upside down, and that certainly should **NOT** alter the radial field.

How about a “circumferential” component? again the answer would be no. For B_ϕ would be constant around an Amperian loop concentric with the solenoid, and therefore:

$$\oint \mathbf{B} \cdot d\mathbf{l} = B_\phi (2\pi s) = \mu_0 I_{\text{enc}} = 0,$$

since the loop encloses no current.

So the magnetic field of an infinite, closely wound solenoid **runs parallel to the axis**. From the right-hand rule, we expect that it points upward inside the solenoid and downward outside. Moreover, it certainly approaches zero as we go very far away. With this in mind, let's apply Ampère's law to the two (2) rectangular loops: Loop 1 lies entirely outside the solenoid, with its sides at distances a and b from the axis:

$$\oint \mathbf{B} \cdot d\mathbf{l} = [B(a) - B(b)] L = \mu_0 I_{\text{enc}} = 0,$$

which means:

$$B(a) = B(b)$$

Evidently the field outside does not depend on the distance from the axis. But we agreed that it goes to zero for large s . It must therefore be zero everywhere! As for Loop 2, which is half inside and half outside, Ampère's law gives

$$\oint \mathbf{B} \cdot d\mathbf{l} = BL = \mu_0 I_{\text{enc}} = \mu_0 nIL,$$

where B is the field inside the solenoid. The right side of the loop contributes nothing, since $B = 0$ out there. We can therefore write:

$$\mathbf{B} = \begin{cases} \mu_0 nI \hat{z}, & \text{inside the solenoid,} \\ 0, & \text{outside the solenoid.} \end{cases}$$

Notice that the field inside is **uniform** given it doesn't depend on the distance from the axis. In this sense



the solenoid is to magnetostatics what the parallel-plate capacitor is to electrostatics as a simple device for producing strong uniform fields ■

Magnetic Field at a Distance Q28

Find the magnetic field a distance s from a long straight wire, carrying a steady current I .

Q28

We know the direction of $\mathbf{B}$ is “circumferential”, circling around the wire as indicated by the right-hand rule. By symmetry, the magnitude of $\mathbf{B}$ is constant around an Amperian loop of radius s , centered on the wire. Therefore Ampère’s law gives

$$\oint \mathbf{B} \cdot d\mathbf{l} = B \oint dl = B2\pi s = \mu_0 I_{\text{enc}} = \mu_0 I,$$

or we can also write it in the form:

$$B = \frac{\mu_0 I}{2\pi s} \quad \blacksquare$$

5

Electrodynamics

An EMF Battery Q29

A battery of Electro-motive Force (EMF) $\mathcal{E}$ and internal resistance r is hooked up to a variable load resistance R . To deliver the maximum possible power to the load, what resistance should you choose.

Note: you can't change $\mathcal{E}$ and r .

Q29

To deliver the maximum possible power:

$$I = \frac{\mathcal{E}}{r + R} \quad \text{A,}$$

$$P = I^2 R = \frac{\mathcal{E}^2 R}{(r + R)^2} \quad \text{W,}$$

$$\frac{dP}{dR} = \mathcal{E}^2 \left[\frac{1}{(r + R)^2} - \frac{2R}{(r + R)^3} \right] = 0,$$

$$r + R = 2R \quad \Omega \rightarrow R = r \quad \Omega \quad \blacksquare$$

Current Carrying Wire Q30

A square loop of wire (side ℓ) lies on a table, a distance s from a very long straight wire, which carries a current I . (**Tip:** The magnetic field of a straight wire is $\mathbf{B} = \mu_0 I / (2\pi s) \hat{\phi}$)

- Find the flux of $\mathbf{B}$ through the loop.
- If someone now pulls the loop directly away from the wire, at speed v , what emf ($\mathcal{E}$) is generated? Also, what direction (clockwise or counter-clockwise) does the current flow?
- What if the loop pulled to the right at speed v , instead of away?

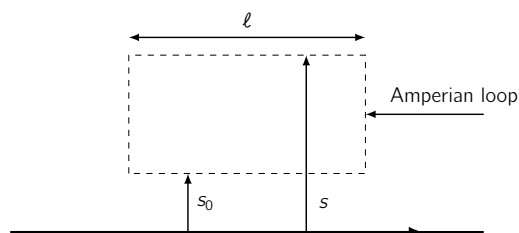


Figure 5.1: Diagram for the question "Current Carrying Wire".

Q30

The solution to the sub questions are as follows.

- The field of long wire is:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi} \quad \text{T}$$

which gives

$$\begin{aligned} \Phi &= \int \mathbf{B} \cdot d\mathbf{a} = \frac{\mu_0 I}{2\pi} \int_a^{s+\ell} \frac{1}{s} (\ell ds) \\ &= \frac{\mu_0 I \ell}{2\pi} \ln \left(\frac{s+\ell}{s} \right) \quad \text{Wb} \quad \blacksquare \end{aligned}$$

- The electromotive force ($\mathcal{E}$) is:

$$\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{\mu_0 I \ell}{2\pi} \frac{d}{dt} \ln \left(\frac{s+\ell}{s} \right) \quad \text{V}$$

we know $ds/dt = v$ therefore:

$$\begin{aligned} \mathcal{E} &= -\frac{\mu_0 I \ell}{2\pi} \left(\frac{1}{s+\ell} \frac{ds}{dt} - \frac{1}{s} \frac{ds}{dt} \right) \\ &= \frac{\mu_0 I \ell^2 v}{2\pi s(s+\ell)} \quad \text{V} \quad \blacksquare \end{aligned}$$

the field points **outwards**, therefore the force on the charge in the nearby side of the square is to the right. In the far side it is also to the right, but here the field is weaker, the current flows **counter-clockwise**.

- This time the flux (i.e., the magnetic field strength won't change) is constant so $\mathcal{E} = 0 \text{ V}$.



The Induced E field of a Wire Q31

An infinitely long straight wire carries a slowly varying current $I(t)$. Determine the induced electric field (E), as a function of the distance s from the wire.

Q31

In the quasi-static approximation, the magnetic field is $rB = \mu I / 2\pi s$ T, and it circles **around the wire**. Similar to the B field of a solenoid, E here **runs parallel to the axis**. For the rectangular *Amperian loop* in the question, Faraday's law provides:

$$\begin{aligned} \oint \mathbf{E} \cdot d\mathbf{l} &= E(s_0)\ell - E(s)\ell \\ &= -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a}, \\ &= -\frac{\mu_0 \ell}{2\pi} \frac{dI}{dt} \int_{s_0}^s \frac{1}{s'} ds' \\ &= -\frac{\mu_0 \ell}{2\pi} \frac{dI}{dt} (\ln s - \ln s_0) \quad \text{V.} \end{aligned}$$

Therefore the electric field is determined to be:

$$\mathbf{E}(s) = \left[\frac{\mu_0}{2\pi} \frac{dI}{dt} \ln s + C \right] \hat{z} \quad \blacksquare$$

where C is a constant of integration.

A Coaxial Solenoid Q32

A short solenoid (length ℓ and radius a with N_1 turns per unit length) lies on the axis of a very long solenoid (radius b , N_2 turns per unit length) as shown below. Current I flows from short solenoid. What is the flux (Φ) through the long solenoid?

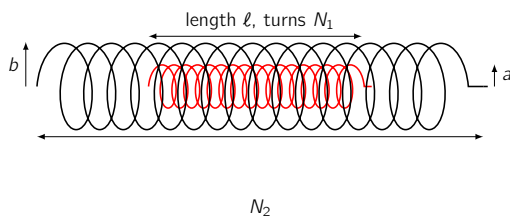


Figure 5.2: Diagram for the question "A Coaxial Solenoid".

Q32

As the inner solenoid is short, it has a rather complicated field; in addition, it puts a different flux through each turn of the outer solenoid. It would be a bit of a difficulty to compute the total flux this way. However, if we approach this situation through *mutual inductance*, the problem become easier. Assume the current is running from the *outer* solenoid, and calculate the flux through the inner one. The field inside the long solenoid is constant:

$$B = \mu_0 N_2 I \quad \text{T},$$

Therefore, the flux through a single loop of the short solenoid is:

$$B\pi a^2 = \mu_0 N_2 I \pi a^2 \quad \text{Wb}.$$

There are $N_1 \ell$ turns in all, so the total flux through the inner solenoid is:

$$\Phi = \mu_0 \pi a^2 N_1 N_2 \ell I \quad \text{Wb} \quad \blacksquare$$

The Self-Inductance of a Toroid Q33

Find the self-inductance of a toroidal coil with rectangular cross section with its inner radius a , outer radius b , height h , which carries a total of N turns.

Tip: the B -field inside the toroid is: $B = \mu_0 NI / 2\pi s$

Q33

The magnetic field inside the toroid is:

$$B = \frac{\mu_0 NI}{2\pi s} \quad \text{T},$$

The flux through a single turn is:

$$\int \mathbf{B} \cdot d\mathbf{a} = \frac{\mu_0 NI}{2\pi} h \int_a^b \frac{1}{s} ds = \frac{\mu_0 NI h}{2\pi} \ln r \frac{b}{a} \quad \text{T},$$

The *total flux* is N times of this value, therefore the self inductance is:

$$L = \frac{\mu_0 N^2 h}{2\pi} \ln \left(\frac{b}{a} \right) \quad \text{H} \quad \blacksquare$$

A Long Coaxial Cable Q34

A long coaxial cable carries current I (the current flows down the surface of the inner cylinder, radius a , and back along the outer cylinder, radius b) is shown below. Using this information, please find the magnetic energy stored in a section of length ℓ . **Tip:** The field between cylinders is $\mathbf{B} = (\mu_0 I / 2\pi s) \hat{\phi}$.



Q34

According to Ampere's law, the field between the cylinders is:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi} \quad \text{T},$$

Elsewhere, the field is zero. Therefore the energy per unit volume is:

$$\frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi s} \right)^2 = \frac{\mu_0 I^2}{8\pi^2 s^2} \quad \text{W}.$$

The energy in a cylindrical shell of length ℓ , radius s , and thickness ds , then, is:

$$\left(\frac{\mu_0 I^2}{8\pi^2 s^2} \right) 2\pi \ell s \, ds = \frac{\mu_0 I^2 \ell}{4\pi} \left(\frac{ds}{s} \right) \quad \text{W}.$$

Integrating from a to b , we have:

$$W = \frac{\mu_0 I^2 \ell}{4\pi} \ln \left(\frac{b}{a} \right) \quad \text{W} \quad \blacksquare$$

Acronyms

E

EMF	An energy transfer to an electric circuit per unit of electric charge, measured in volts. Devices called electrical transducers provide an EMF by converting other forms of energy into electrical energy. Other types of electrical equipment also produce an emf, such as batteries, which convert chemical energy, and generators, which convert mechanical energy. This energy conversion is achieved by physical forces applying physical work on electric charges. However, electromotive force itself is not a physical force
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O

ODE	Ordinary Differential Equation
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P

PDE	Partial Differential Equation
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